

Meta Open Rack Frame V3 Specification

Revision 1.3

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1.2 Acknowledgements

The Contributors of this Specification would like to acknowledge the following contributors and companies for their input and help developing the design and specification:

Harsha Bojja
Julia Huynh - Meta
Ben Kim - Meta
Saurabh Kulkarni - Meta
Steven Moore - Rittal
Thomas Mortlock
Sriram Srinivasan - Rittal

2. Compliance with OCP Tenets

Please describe how this Specification complies to the following OCP tenets. Compliance is required for at least three of the four tenets. The ideals behind open sourcing stipulate that everyone benefits when we share and work together. Any open source project is designed to promote sharing of design elements with peers and to help them understand and adopt those contributions. There is no purpose in sharing if all parties aren't aligned with that philosophy. The IC will look beyond the contribution for evidence that the contributor is aligned with this philosophy. The contributor actions, past and present, are evidence of alignment and conviction to all the tenets.

2.1. Openness

This specification builds on the base Open Rack and provides an example of a detailed implementation for the community to leverage and promote the building of open products. Detailed design specifications and design documentation needed to build this Meta version of the rack and accessories are incorporated into this specification.

2.2. Efficiency

The sharing of this specification will allow others to leverage and learn from the development work that Rittal & Meta have put into developing a detailed design of the frame and accessories designs.

2.3. Impact

This version of Open Rack provides a detailed design contribution that will allow others to reference in future contributions or designs. This will also allow adopters to leverage all the development work and resulting design to potentially develop and deploy solutions with quicker time to market.

2.4. Scale

The various construction and design requirements outlined were all developed to support the build at Scale of this Meta version frame. By sharing this implementation, the community will see a detailed design that can be leveraged to make the frame and related accessories at scale.

2.5. Sustainability

The frame design has application in multiple IT industry domains and, as a result, has the potential to enable more rack reuse in these different areas.

3. Revision Table

Date	Revision #	Author	Description
03/08/23	0.1	Charest	Draft shared w/ R&P
04/05/23	1.0	Charest	Misc. updates
07/09/23	1.1	Charest	Misc. updates & updates to Fig 7.4.1. 7.5.1, & 7.5.2 . other minor changes.
02/28/24	1.2	Huynh	Design collateral (2D dwgs and 3D CAD updated) to reflect mfg. tolerances and capability or corrections reordered portions of section 7; added in 10 deg ramp test req. in 7.8
06/03/24	1.3	Charest, Daniels	Added Blind mate section 12.15 & section 14, updated Fig. 7.4.1

4. Scope

This document defines technical specifications for a specific implementation, the Meta Version, of Open Rack Frame V3 used in Open Compute Project. This document, along with the Open Rack V3 Base Specification shall comprise product's technical specification. This specification is a supplement to the Open Rack V3 Base Specification specification.

5. Overview

Open Rack is a fundamental building block for the Open Compute ecosystem. It provides the structure and power for interoperability of Open Rack based IT Systems that are contributed to the Open Compute Foundation. This frame design is an implementation of Option 1 outlined in Open Rack V3 Base specification.

Any statements using:

SHALL must meet this requirement to comply with the specification

SHOULD are recommended to meet this requirement but are not required

MAY are suggested if the rack supplier is interested in meeting for interoperability between solutions

6. Physical Specifications

This section provides a high level description of the frame size, supported weight, and capacity. More detailed information is provided in section 7.

6.1 Dimensions

6.1.1 Physical Definition:

The basic frame dimensions are defined below.

6.1.2 Height: (nominal): 2286 mm (90 inches),
floor to frame top

6.1.3 Width (nominal): 600 mm (23.62 inches)

6.1.4 Depth (nominal): 1068 mm (42.05 inches)



6.2 Load Rating

Load rating shall be 1400 kgs (3080 lbs). This does NOT include the mass of the rack. Any IT shelves, doors, side panels and any other accessory items shall be included in the max load.

A cross brace 1-OpenU in height may be added to the rack assembly to meet the load rating for configurations heavier than 800kgs (1763 lbs). For racks that require the cross-brace, the recommended range for the brace locations is rack positions 18OU to 27OU, with the default location being position 23OU.

6.3 OU/RU Capacity

This frame design has the capacity to support 44 OpenU or 47 RU of IT gear. The rack design also shall have the ability to support 88 - ½ OpenU IT gear as well. The mixing of the various types is possible as the rack design has cut outs for gear support brackets for each option. Note that the spacing and support brackets designs vary and the details regarding the mixing IT gear types is not covered in this specification. Note that a ½ OpenU design guide does not exist, but the features are included in the rack to allow for design flexibility for developers.

7. Mechanical

7.4 Frame X-Section:

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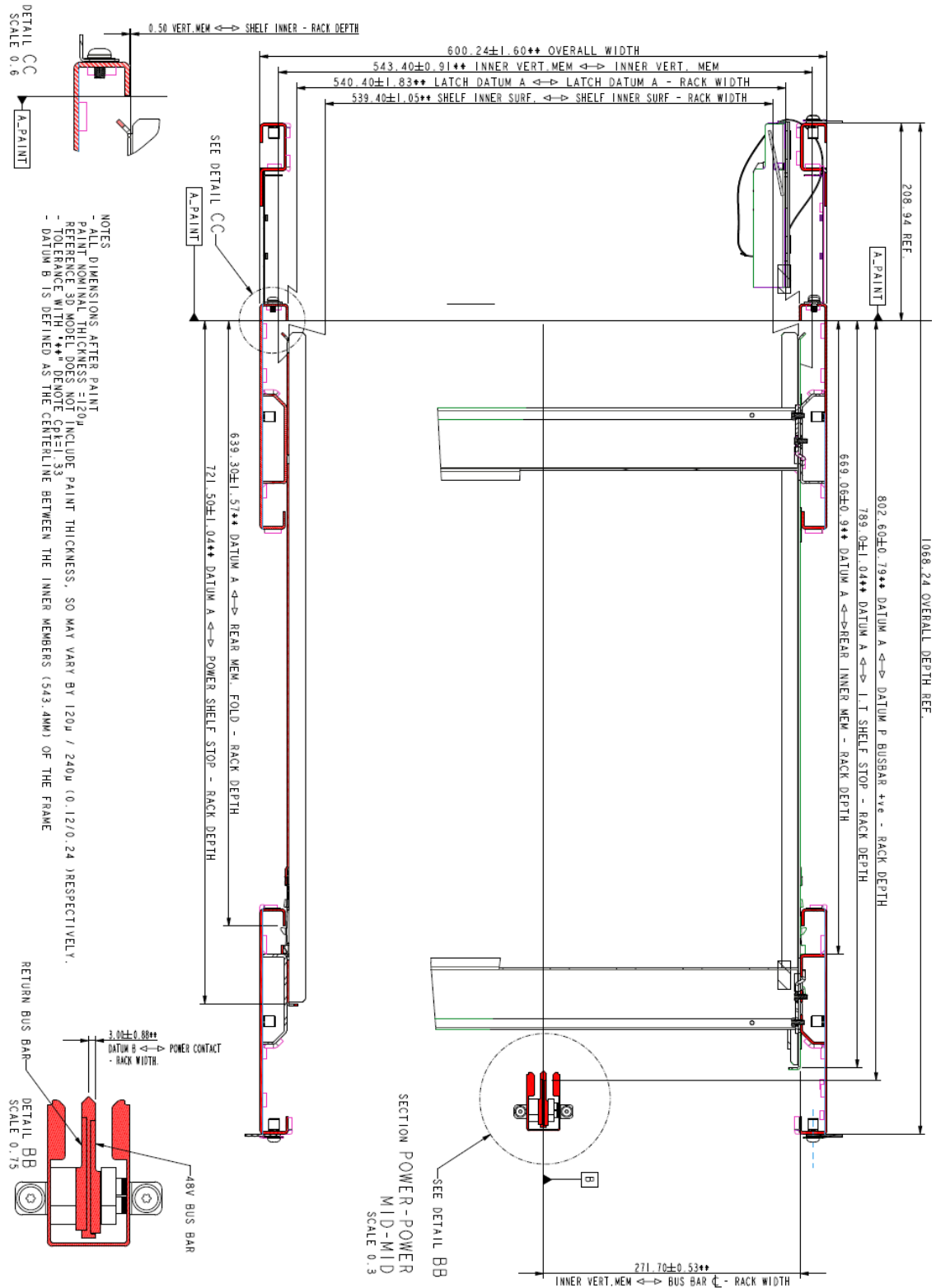


Figure 7.4.1 Frame Cross Section

7.5 Frame Mounting Points

Mounting point locations on the front and rear of the rack are outlined below. These locations can be used for mounting doors or other accessories. There are also mounting points in the sides of the rack. There are threaded inserts in these locations and the details on the thread type can be determined from bill of materials in the top level drawing number C30001; refer to this drawing in the provided collateral.

7.5.1 The fasteners shall be M8 fasteners shall be M8 x 1.25p - 6H

7.5.2 The fasteners shall be M8 fasteners shall be M12 x 1.75p - 6H

7.5.3 The minimum torque out for the M8 rack fasteners is 20N-m.

7.5.4 The minimum torque out for the M12 rack fasteners is 20N-m.

7.5.5 The fasteners shall not protrude above of the frame installed surface

7.5.6 The distance from the frame face to the start of the threads on the M8 fastener shall be 6mm maximum and on the M12 fastener shall be 8mm maximum

7.5.7 The M8 fastener shall have a minimum 7 x full threads.
The M12 fastener shall have a minimum 9 x full threads.

7.5.1 Front Mounting points

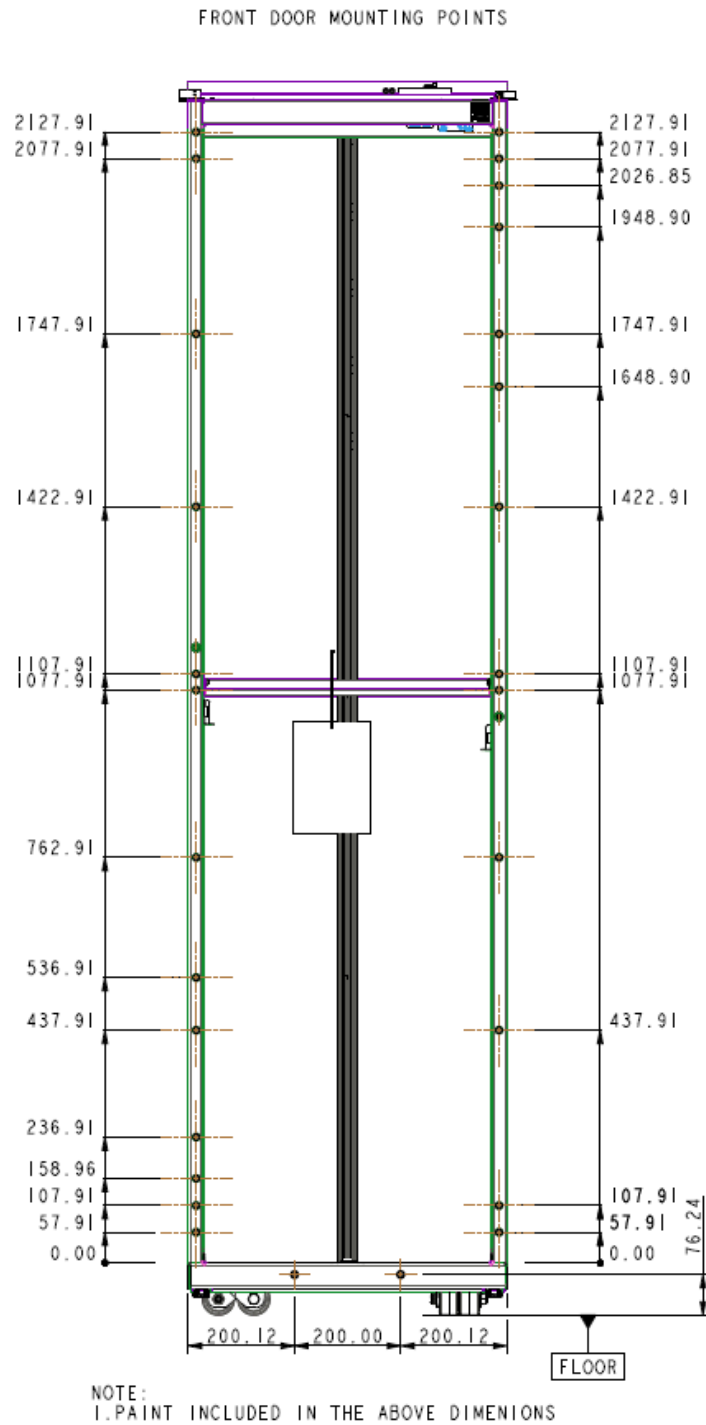


Figure 7.5.1 Front Frame Mounting Points - All M8

7.5.2 Rear Frame Mounting points

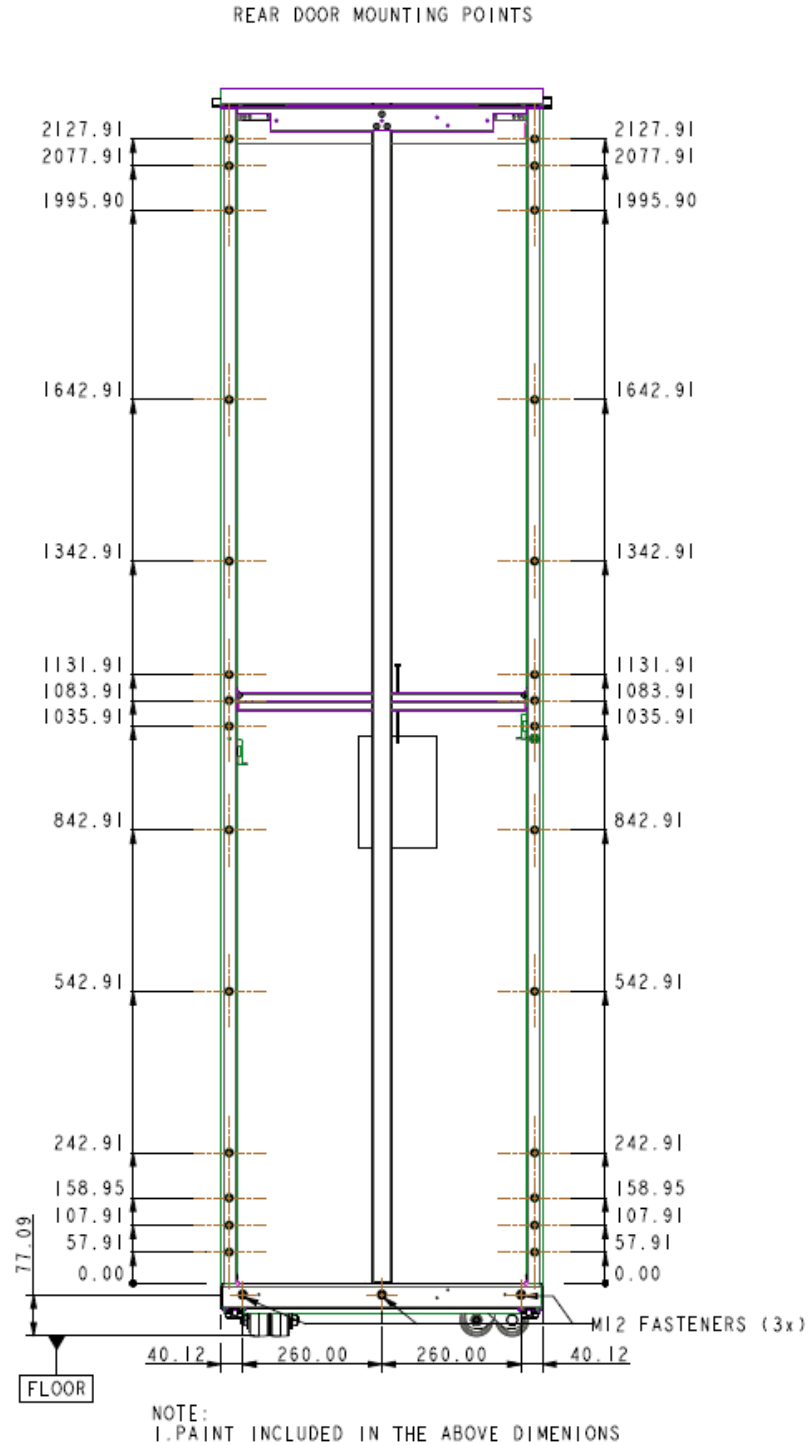


Figure 7.5.2 Rear Frame Mounting Points- All M8, unless otherwise stated

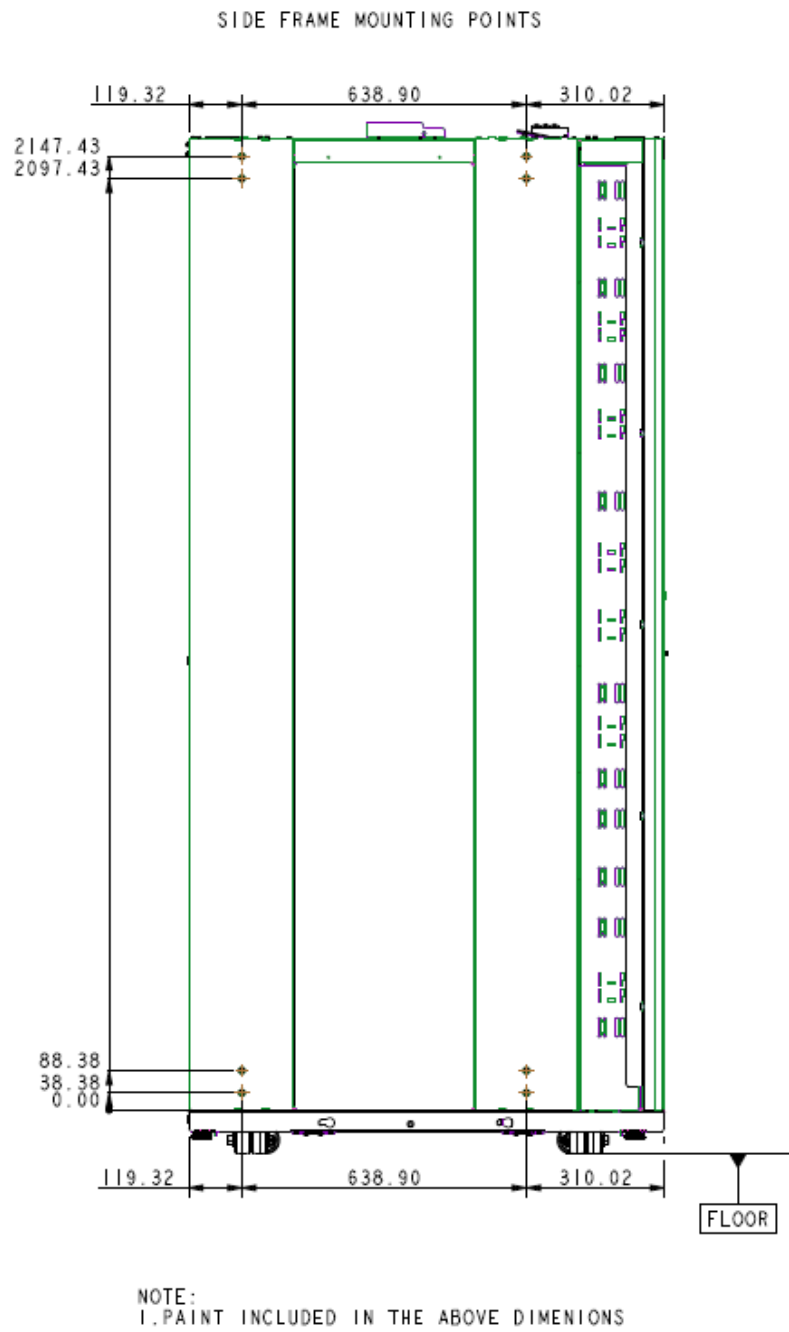


Figure 7.5.3 Side Frame Mounting Points - All M8

7.6 Air Flow

The rack is designed with front to back airflow. Under normal operating conditions, no openings in the rack sides, top, or bottom are allowed where hot air can recirculate to the cold aisle of the data center.

7.7 Cabling

7.7.1 Rack shall provide an exit hole in the bottom of the rack 25mm x15mm minimum in size under each cable zone in the front of the rack to allow for a data cable under the floor to pass to the cable zone.

7.7.2 To prevent cable damage, sharp edges and burrs must be removed around cable routing areas. For areas where sharp edges cannot be removed, proper protection or guides must be put in place to ensure cables do not make contact with the sharp edges.

7.7.3 Cables can be added and removed from the cable management system without tools.

7.7.4 Provides a cable pass through the top and the bottom of the rack frame for four power cables. Max dimensions of power cable connector housing is 68mm x 24mm.

7.7.5 Rack shall have openings at the top of the rack for cable pass through as shown in figure(s) below.

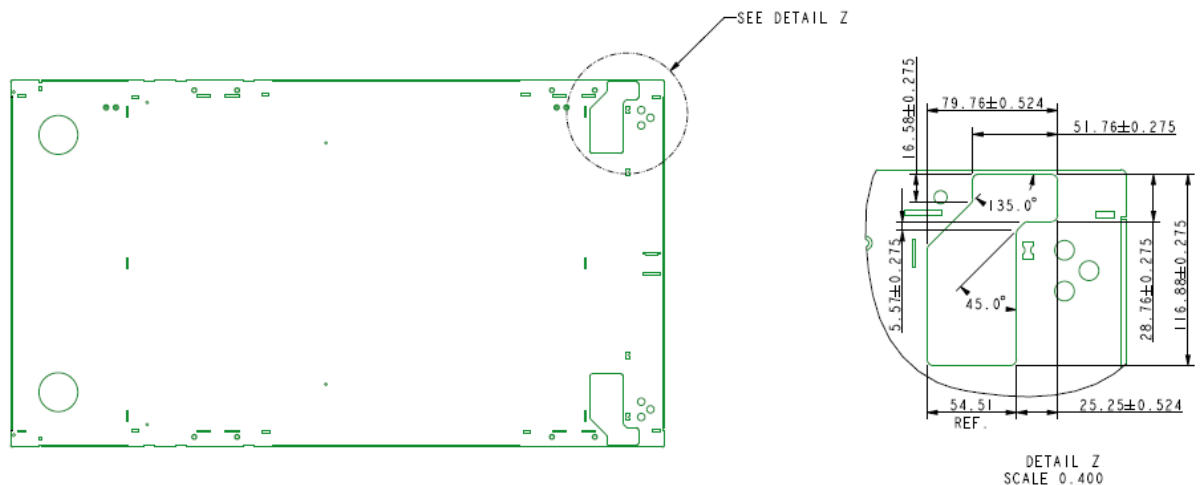


Figure 7.7.5.1 Top of rack cable openings WITHOUT plastic grommets installed

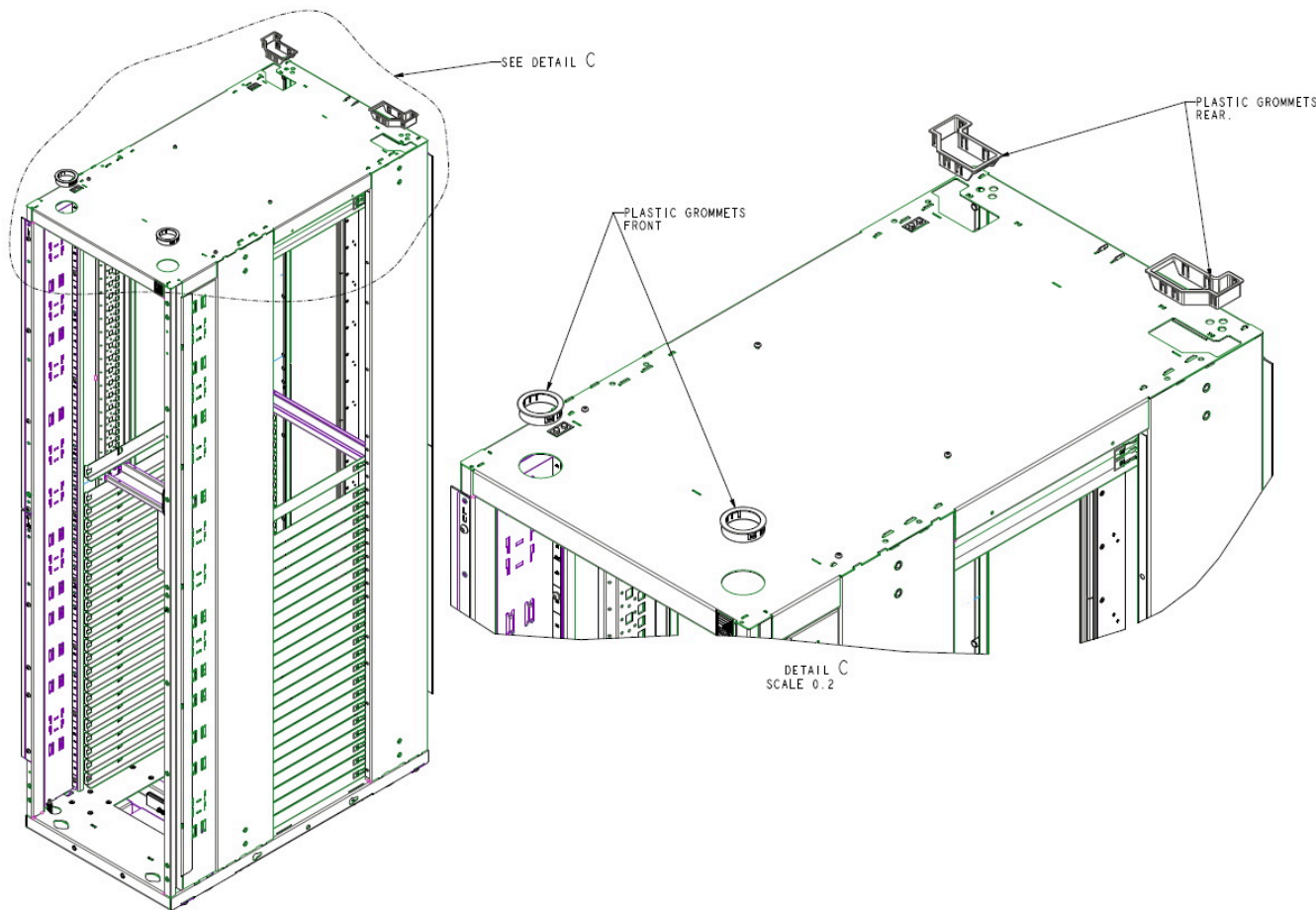


Figure 7.7.5.2 Top of rack cable openings showing where rear plastic grommets install

7.8 Rolling Movement - Ramp Test Requirements

The rack shall be able to transition a 10-degree ramp both up and down. From a flat surface, traverse a 10 degree ramp up to an elevated flat surface . From an elevated flat surface, traverse down a 10 degree ramp down to a flat surface. Each surface SHALL be large enough for the entire rack (the full frame depth & width) to roll onto prior to transitioning to the next surface. This test should be performed w/ the rack in two different orientations: 1) rack front facing towards the ramp and 2) rack rear facing towards the ramp.

7.9 Rack Mover Related Requirements

The frame shall follow the frame profile and design as outlined in the drawings and 3D CAD as the interfaces for the rack mover require them to constrain and move the rack.

Frame Pull Test: Restrain rack at rear using the bottom most Qty. 4 - M8 screws, 2 per side, in the rear verticals of the frame. Attach two pull straps with hooks to the rack, one each to the front left

and front right members with the hook grabbing the outer slots, locate the straps 582mm off the floor. Connect the 2 straps at the end opposite the rack and apply a total minimum load of 900N to the frame through both straps at the same time (450N per strap) and release. Verify no permanent damage or deformation. See Figure 7.8.1 below for reference.



Figure 7.9.1: Image showing 2 straps with hooks attached to rack during test.

8.0 Prescribed Materials

All critical frame & accessory materials are called out in detail within the respective documentation provided with this specification (in separate drawings).

9.0 Environmental Requirements

- 9.1 Operating Temperature: 10°C to 60°C
- 9.2 Long-term Storage: -40°C to 50°C and 5-95% RH
- 9.3 Short-term Storage: -20°C to 65°C and 10-80% RH
- 9.4 Operating Humidity: 85%max, 5°C dew point minimum

10.0 Color & Markings

10.1 Color

All painted metal components to be black and applied at a thickness of 0.12mm +/-10%.

10.2 U Markings

All “U” markings shall be marked per requirements in the frame base specification. The locations for the markings are outlined in the rack label drawings provided.

11.0 Electrical Grounding

11.1 Telco Ground Lug:

11.1.1 Provide M5X0.8 nut features as an electrical ground to mate with Data Center Telco Ground Lug as defined in FB PN: 18-000233.

11.1.2 Conductivity of the ground path shall be protected from rust and corrosion over the life of the product.

11.1.3 Green ground screws shall be provided with the rack and have a hex head 8mm across the flats.

11.1.4 Ground points shall be located per 3D CAD on the top surface of the rack for overhead grounding and the rear base surface to support underfloor grounding.

11.2 IT Gear Ground Path

11.2.1 Rack shall provide an electrically conductive path from the IT equipment in the rack to the rack grounding lug on the top of the rack. For the Meta implementation of ORv3, the ground path goes from the busbar → the frame → the ground lug.

11.2.2 This path shall not pass through any surfaces that are not protected from rust and corrosion such as un-plated surfaces.

11.2.3 Conductivity of the ground path shall be protected from rust and corrosion over the life of the product.

11.3 Ground Strap Connection

11.3.1 Ground Strap Connection (Banana plug) located at the front of the rack.

11.3.2 Rack shall provide a location along the front of the rack that does not interfere with the removal of IT service gear

11.3.2 Ground point will have a contrasting color to help visually identify the location

12.0 Optional Accessories & Kits:

General Requirements applying to all SKUs: all interfaces between the frame and components listed below shall be adequately defined to ensure proper fitment and/or performance.

12.1 Caster FRU:

12.1.1 The FRU will contain a single caster assembly and hardware.

12.2 Leveler FRU

12.2.1 The FRU will contain a single leveling foot assembly and hardware.

12.3 Side Panels for Air Flow Containment

12.3.1 Shall stay installed during the rolling movement test

12.3.2 Shall require only 1 person to install and remove

12.3.3 Shall be designed with hardware that features T30 Torx heads

12.4 Cable Management tray

12.4.1 Shall be designed as a removable kit.

12.4.2 May be installed using tools.

12.4.3 Shall provide 3090 mm² of cross-sectional area per side.

12.5 Busbar assembly

12.5.1 Complies with OCP ORV3 Meta Busbar Specification.

12.6 Air Containment Filler Blanks

This accessory is not a Rittal provided design, CAD & drawing are provided for reference.

12.6.1 OpenU Air Filler Blanks are to be used to block any unused OpenU positions within the rack to prevent airflow bypass.

12.6.2 Design of OpenU Air Filler Blanks (various sizes) shall feature tool-less installation and may utilize the IT gear shelves for mounting.

12.7 Air Containment from Rack Frame to Facility Hot-Aisle Containment

12.7.1 Shall provide a method to extend the height of the rack to meet the data center containment at a height of 2413mm (95") from the floor.

12.7.2 Shall ensure that air cannot pass under the bottom of the rack under normal conditions.

12.7.3 Frame shall provide 25mm of flat space on top of the rack canopy on the cold aisle side for mounting a cold-aisle containment panel.

12.8 IT Support Bracket

The IT gear will optionally sit on a set of brackets that form a shelf. CAD & drawing included in specification zip file for reference.

12.8.1 Shall provide a method of toolless mounting IT support brackets directly to the rack frame.

12.8.2 Shall be no more than 1-OpenU tall and serviceable without removing or interfering with other IT support brackets or nearby IT chassis.

12.8.3 Must be able to install or remove from inside the rack frame.

12.8.4 Shall support 80 kg per set under dynamic loads without taking any permanent deformation. Shorter versions of the bracket can be rated at a lower weight (like brackets for the shorter depth power shelf).

12.8.5 Static Load at end: Support 50 kg static load applied over 25mm of leading edge of a pair of the support brackets installed in the rack without taking any permanent deformation.

12.8.6 Dynamic Load at end: 32kg dropped from 30mm height applied over 25mm of leading edge of a pair of the support brackets installed in the rack without taking any permanent deformation, 1 drop

12.8.7 Bracket Pull Out Force: rack bracket installed in rack, pull on rack towards front of rack. The retention latch shall have a 400N minimum pull out force. Test with both left and right versions as well as any rail design variations.

Pull force to be applied with following criteria (reference Figure 12.8.7.1 below):

- Perpendicular to the front face in the vertical and Horizontal direction.
- Apply force in a controlled manner, possibly using a ratchet-strap attached to a fixed structure.
- The front horizontal surface of the shelf may be drilled to attach the pulling apparatus directly to the horizontal surface flange of the bracket.

See Figure 12.8.7.2 Example of Bracket pull out force testing

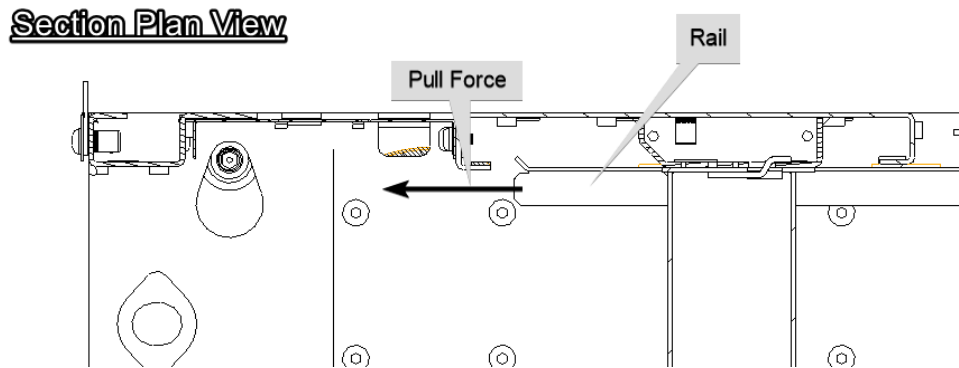


Figure 12.8.7.1 Partial Top Section View of Rack



Figure 12.8.7.2 Image showing Bracket pull out force testing

12.9 Power Shelf Support Bracket

The power shelf will optionally sit on a set of brackets that form a shelf. CAD & drawing included in specification zip file for reference.

12.9.1 Shall provide a method of toolless mounting IT support brackets that mount directly to the rack frame.

12.9.2 Shall be no more than 1-OpenU tall and serviceable without removing or interfering with other IT support brackets or nearby IT chassis.

12.9.3 Must be able to install or remove from inside the rack frame.

12.9.5 Static Load at end: Support 50 kg static load applied over 25mm of leading edge of a pair of the support brackets installed in the rack without taking any permanent deformation.

12.9.6 Dynamic Load at end: 32kg dropped from 30mm height applied over 25mm of leading edge of a pair of the support brackets installed in the rack without taking any permanent deformation,, 1 drop

12.9.7 Bracket Pull Out Force: rack bracket installed in rack, pull on rack towards front of rack. The retention latch shall have a 400N minimum pull out force. Test with both left and right versions as well as any rail design variations. Refer to section 12.8.7 for details.

12.10 Cross Brace

The rack may include a cross brace for configurations. CAD & drawing included in specification zip file for reference.

12.10.1 Cross-brace kit shall include all hardware required for installation.

12.10.2 T25 Torx driver may be used for installation.

12.10.3 IT gear may be removed from the rack to install or remove the brace.

12.10.4 Cross brace shall require 1-OpenU in height maximum.

12.10.5 For configurations requiring a cross-brace, the rack shall have features to support brace locations in positions OpenU 18 to OpenU 27.

12.11 Front and Rear Door

12.11.1 The door could be EMI, security, or heat exchanger and is out of the scope of this document.

12.11.2 The frame design shall enable both a front and rear door. Rear door shall be split.

12.11.3 Rear door and/or spacer combined weight shall not exceed 150 kg as set by the requirement of the frame to support a rear door heat exchanger.

12.11.4 Door mounts (such as hinges) shall support the door weight under dynamic loads.

12.11.5 Door swing angle must be >90° without door frame extending outside the width of the rack frame (so it does not interfere with adjacent racks). It is acceptable for handles/locks to extend beyond the rack width when the door is opened.

12.11.6 Any optional doors are not included in the rack depth limits defined Section 6.1 of this specification.

12.12 Stabilizer

Some configurations of racks may require a stabilizer kit for rack stability. The stabilizer shall meet the following requirements:

12.12.1 Installation and removal without tools

12.12.2 Support the rack at the 10 degree tip test at maximum weight listed in this specification

12.12.3 Left and right stabilizers are the same design (i.e. not left/right handed)

12.12.4 Work with frame interface features defined in Open Rack V3 Base Specification

12.13 Tool less Front (Plastic) Cable Clip

This accessory is not a Rittal provided design, CAD & drawing are provided for reference. The cable clip shall meet the following requirements (detailed design information will be provided in the engineering documentation):

12.13.1 Installation into front cable manager without tools.

12.13.2 Removal from both inner and outer sides of front cable manager without tools.

12.13.2 Design so that can install forward or rearward facing.

12.13.3 Part shall meet dimensions and tolerances in the design documentation.

12.13.4 Part to be molded in green and per material outlined in drawing.

12.14 Tool less Busbar Access Cover

This accessory is not a Rittal provided design, CAD & drawing are provided for reference. The cable clip shall meet the following requirements (detailed design information will be provided in the engineering documentation):

12.14.1 Shall be a tool less design

12.14.2 Part shall grip onto the busbar in a sturdy manner and not drop from gravity.

12.14.3 Part shall meet dimensions and tolerances in the design documentation.

12.13.4 Part to be molded in green and per material outlined in drawing.

12.15 Liquid Cooling Kit

This kit is an add on kit to support blind mate liquid cooling. Details regarding this kit are discussed in detail in Section 14.

13. Environmental and Regulations

Below are requirements listed in addition to requirements listed in the Open Rack V3 Base Specification.

13.1 WEEE Directive (2012/19/EU); aims to reduce the environmental impact of EEE by restricting the use of certain substances during manufacture.

13.2 US EPA - **Substances of Concern** In articles as such or in complex objects (Products) The vendor shall provide the declaration of compliance document stating that a product doesn't contain any substances regulated by EPA 40 CFR751

14. Blind Mate Liquid Cooling

One of the enhancements for Open Rack V3 (ORv3) over the previous rack is the support for liquid cooling. The blind mate solution developed in the OCP ORv3 Blind mate interfaces group is based on the ORv3 frame. Blind mate liquid cooling reduces service times and can have layout advantages vs. manual connections. For blind mate racks, an added liquid cooling kit is added with interface brackets and stiffeners; the kit is designed to be a bolt-on kit added to a standard Meta Version ORv3 Frame. This specification outlines the various requirements for the ORv3 liquid cooled rack as well as providing some background on the overall rack solution design considerations for developers.

The solution also includes the (BMQC) blind mate valves, the blind mate manifolds, and the BMQC specific IT gear chassis reference design. The interfaces and design information for the blind mate valves (BMQC), IT gear, and manifolds are each defined in their respective specifications or guidelines elsewhere. They include:

- *Open Rack V3 Base Specification*
- *Meta Open Rack Frame V3 Specification (this document)*
- *Open Rack V3 IT Gear Design Guide*
- *Open Rack V3 Blind Mate Quick Connector Specification (BMQC)*
- *Open Rack V3 Blind Mate Manifold Base Specification*

14.1 Mechanical Overview

14.1.1 Basics

Frame: Figure 14.1.1.1 below shows an illustration of the reference design used during the development of the ORv3 blind mate hardware. Other designs & layouts are

possible, but this is the specific layout used by this CLA group. Some of the key requirements for the ORv3 frame and liquid cooling kit include:

- Structural strength to counteract valve spring and fluid forces between the manifolds and chassis.
- Maintaining critical dimensions and tolerances at interfaces.
- The liquid cooling kit shall be an add-on kit installed with common tools (not part of the weldment) to allow one to easily convert a standard rack to a blind mate liquid cooled rack.
- Solution is based on the Meta Version of the Orv3 frame (specification available on OCP website).

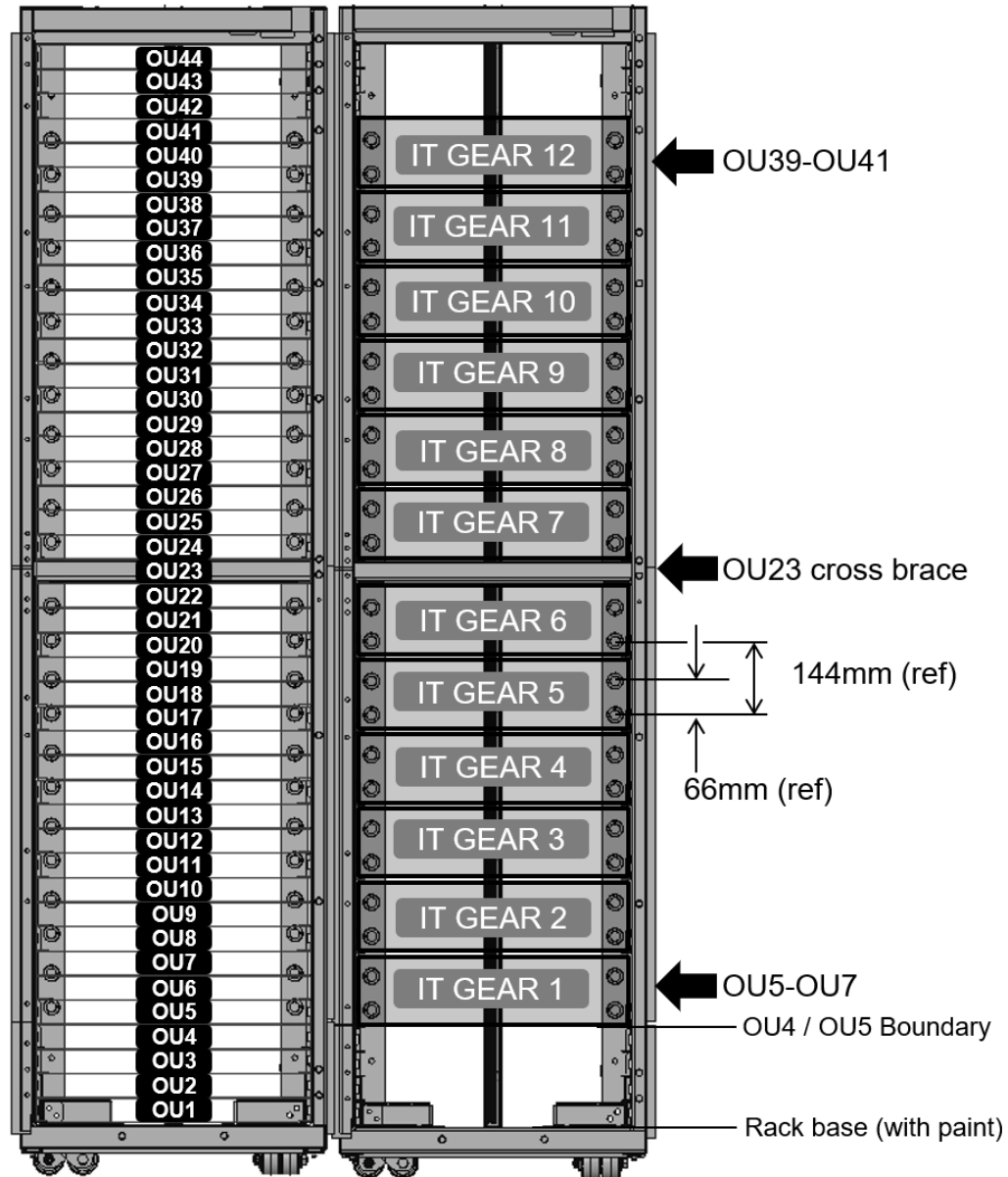


Fig. 14.1.1.1 - Reference ORv3 Blind Mate Rack Layout

Figure 14.1.1.2 and Figure 14.1.1.3 below show critical interface dimensions between the rack frame and manifold. The location, tolerance and, if required, deflection for the components required for the critical dimensions below are covered in their respective specifications and guidelines.

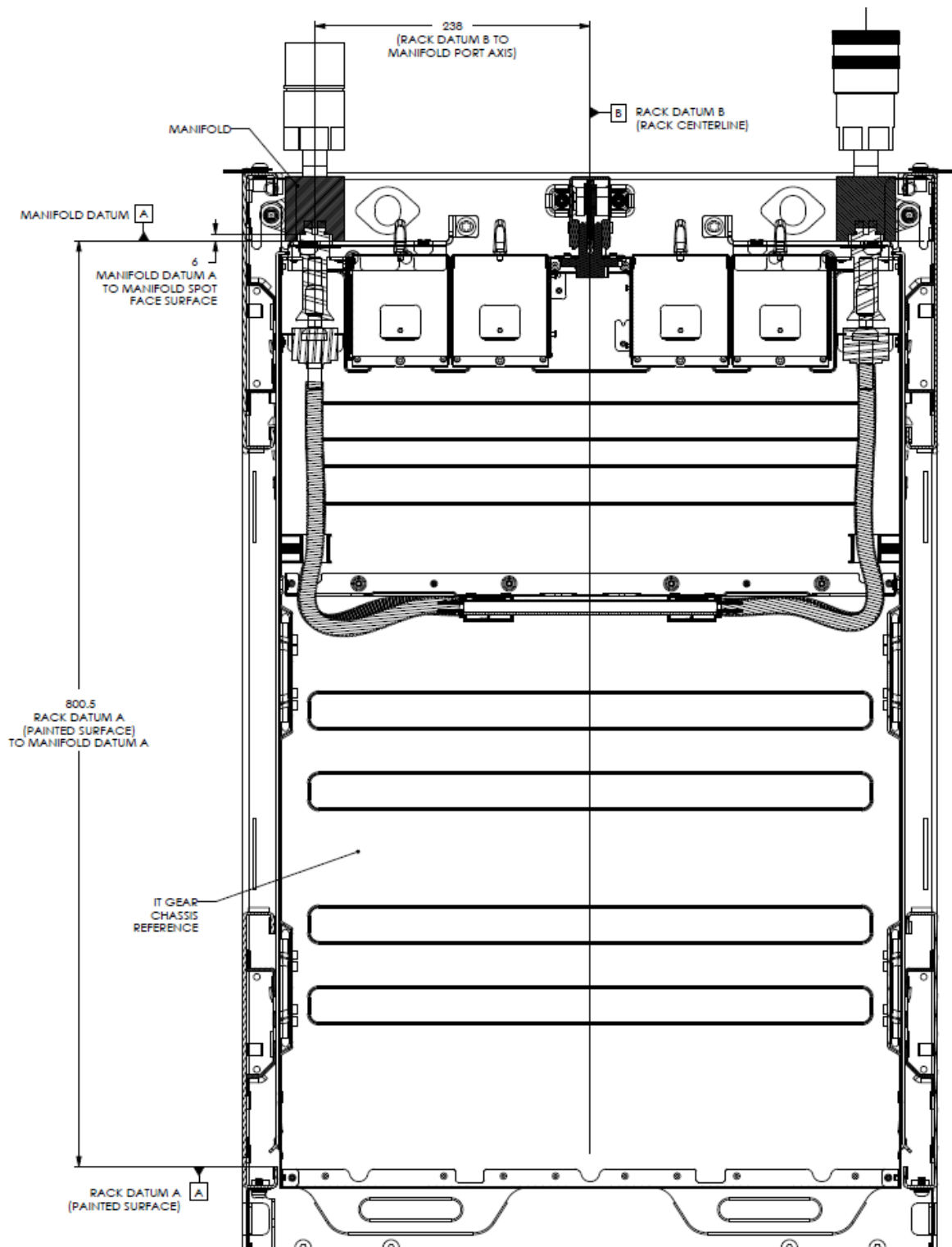


Fig. 14.1.1.2 - Critical horizontal dimensions for rack to manifold interfaces.

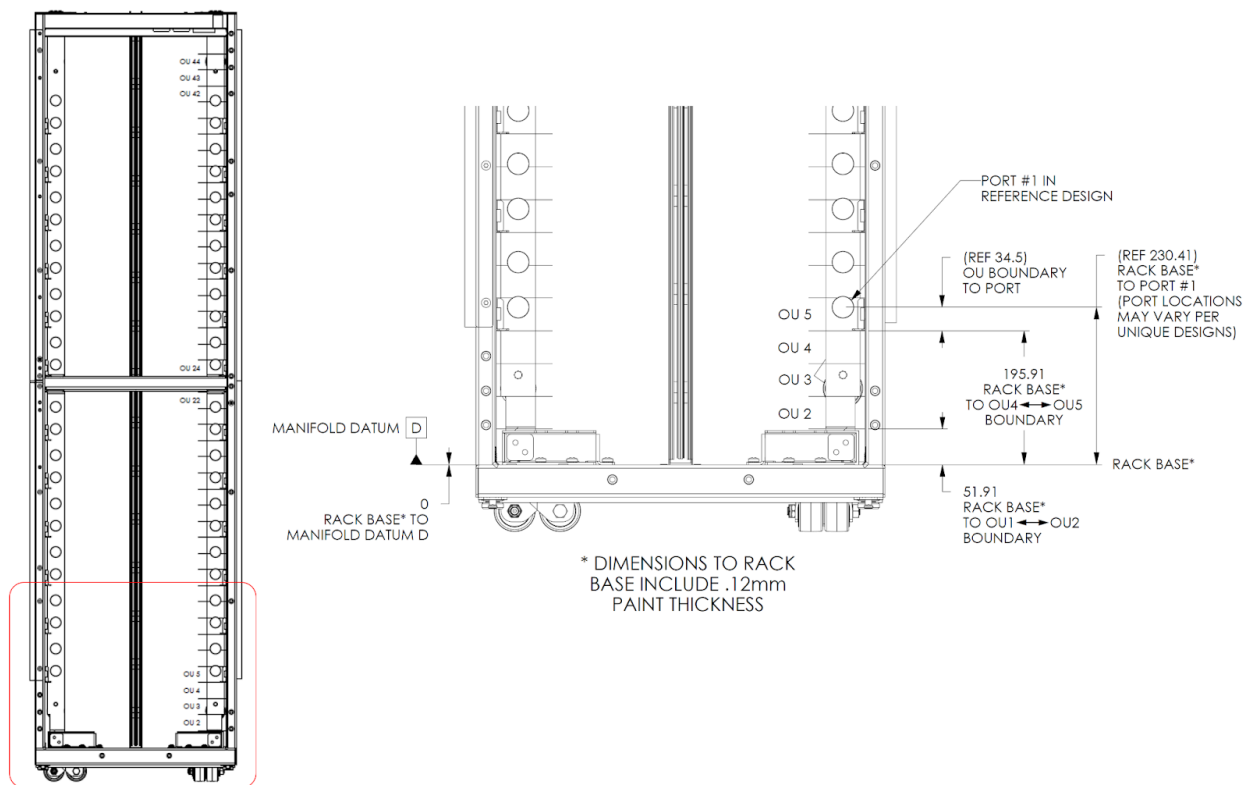


Fig. 14.1.1.3 - Critical vertical dimensions for rack to manifold interfaces.

14.1.2 Collateral

Information on the drawings and 3D CAD models that are included with the specification:

- Reference design 3D CAD files with the rack frame with liquid cooling kit, blind mate manifolds, BMQC valves, and the reference 3OU blind mate liquid cooling chassis are provided in the design collateral.
- A 3D CAD assembly file and 2D drawing of the key interface dimensions and tolerances for the rack with just the liquid cooling kit installed is also provided with the collateral.
- The frame and accessories are all manufactured with a 0.12 mm of powder coat added thickness. However, in the 3D CAD collateral, the frame assembly is not modeled with the powder coat thickness. The liquid cooling kit components are modeled with the powder coat thickness; keep this in mind when looking at the CAD or performing any analysis or design work.
- The blind mate liquid cooled chassis provided is a simplified version of the group's development chassis and is a reference design. Detailed 3D CAD and a drawing of key blind mate chassis interfaces and tolerances are provided in the *Open Rack V3 IT Gear Design Guide*.

- The manifolds in the model are simplified reference models, not an actual design, that outline the interfaces and design basics. Refer to the *Open Rack V3 Blind Mate Manifold Base Specification* for details on the manifold design and requirements.
- The blind mate valve (BMQC) CAD models provided are simplified models, refer to the *Open Rack V3 Blind Mate Quick Connector Specification (BMQC)* for more details.
- For general rack tolerance and design information refer to the *Open Rack V3 Base Specification*.

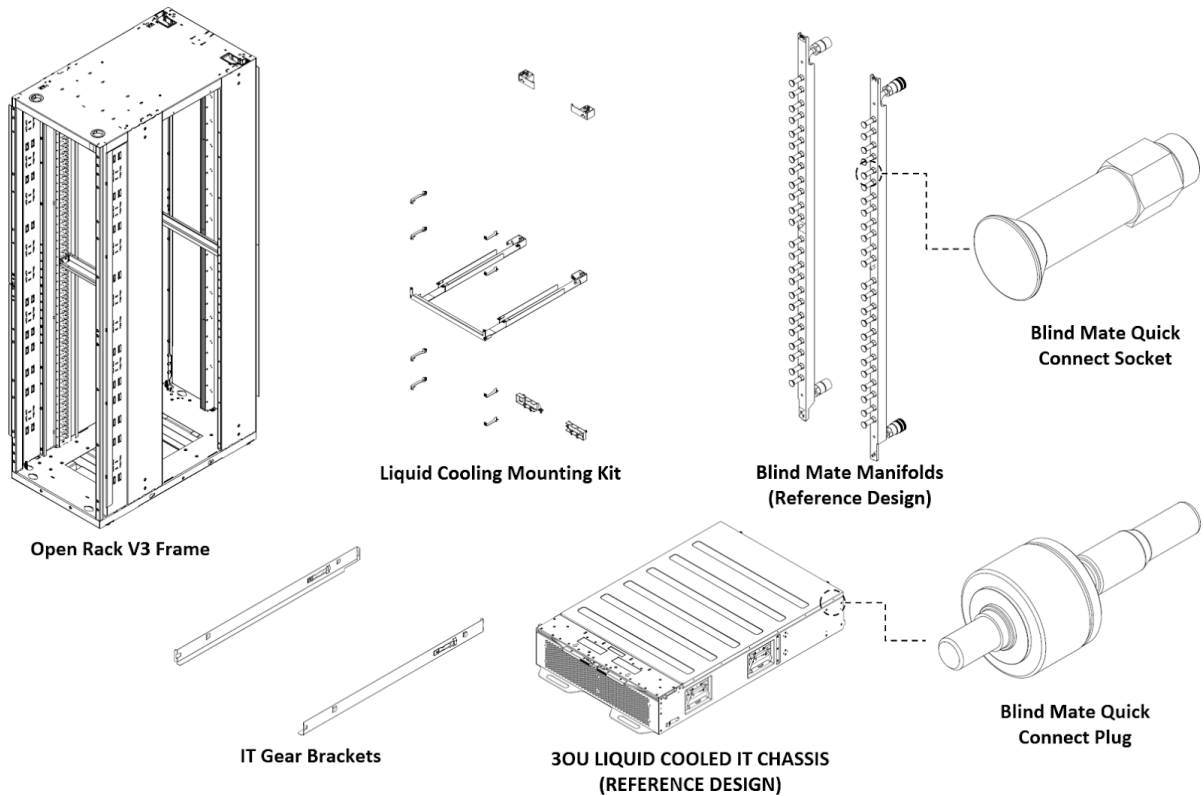


Fig. 14.1.2.1 - 3D CAD model reference

14.2 Overall Design Considerations:

14.2.1 Reference Rack Configuration

The 12 chassis rack configuration provided is for reference, each solution may require unique solutions. The reference layout used for development had 24 valves per manifold and 12 x 30U IT gear chassis, split by a cross brace at OU23. Other layouts are possible, and it is up to the developer to design and evaluate their solution. The CLA group developed the various components & fundamental interface requirements to allow

developers to create their own solutions. It is up to the developer to design, develop, and test to validate their designs. Some of the mechanical design considerations for solution developers to consider when developing solutions are provided below. To ensure the information for a particular component discussed is accurate, the developer is directed to the component specifications for that component for the appropriate dimensions and requirements in some cases below.

14.2.2 Valve Working Range

Referencing the various interfaces (valves, IT gear, manifolds, & rack), the blind mate rack assembly level reference design was developed with a nominal valve mate distance (provided in Blind Mate Valve Specification Section 9.3). This distance assumes the chassis injectors are flush with the rack IT gear catch edge Datum A and lies between the maximum and minimum mate distances, with the distance being based to be further from the maximum mate distance..

To ensure reliable performance, it is critical that the design implementation prevents the valves from exceeding the maximum mate distance listed in Section 9.3 Working Range of the *Open Rack V3 Blind Mate Quick Connector Specification*. It is recommended that developers ensure their solutions have adequate design margin to prevent this. Exceeding this maximum mate distance will result in fluid flow performance issues as the valves will start to close; the rate at which they close when exceeding this mate distance will vary and is not controlled in the specification.

It is also important to verify the solution ensures the valve mate distance is never less than the minimum value listed in Section 9.3 of the *Open Rack V3 Blind Mate Quick Connector Specification*. In the reference design, there is a 1.5mm gap between the rear chassis stop features and the IT bracket stop features when the chassis injectors are flush with Datum A catch edge. See Figure 14.2.2.1 below for details. This gap is required to account for tolerances between the IT gear chassis and rack interfaces and ensure fit. This gap means that the IT gear chassis can shift rearward towards the manifolds by this distance (1.5mm) and keep this in mind when designing to ensure that the minimum valve mate distance is not violated. Violating this distance could result in binding and/or irreparable damage to the valves.

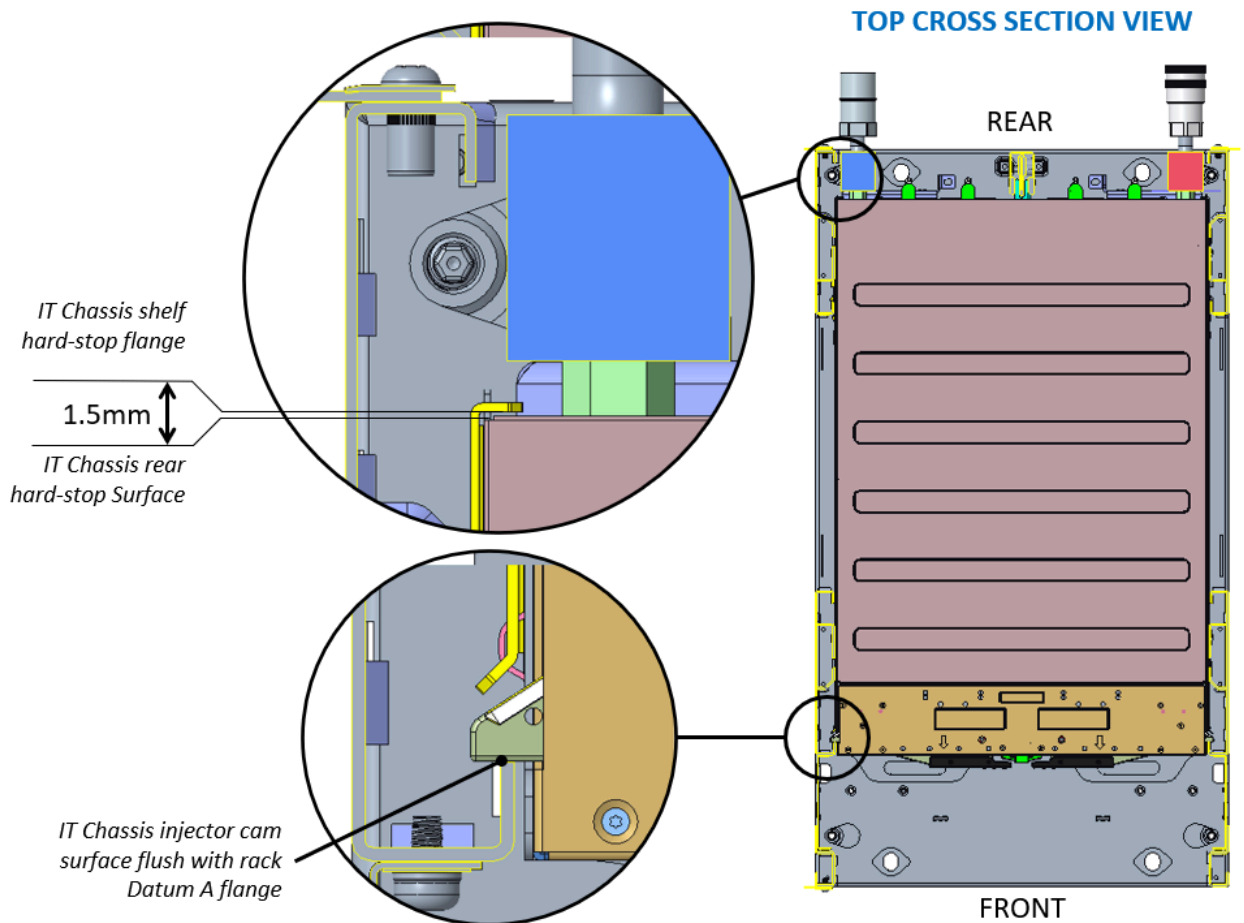


Fig. 14.2.2.1 - Top cross-section view showing gap between IT shelf hard stop and chassis

Critical factors that can affect the working range and performance include, but are not limited to, tolerances of the various subcomponents & interfaces (rack, manifolds, chassis, IT gear support brackets, etc.), manufacturing tolerances, gaps, skew between components due to gaps, deflections due to loads from the valves and coolant pressure (deflections at the rack, manifolds, and chassis), chassis loading & friction, shipping and handling, and CTE (coefficient of thermal expansion) effects as the ambient and or fluid temperatures change. Appropriate structural analysis, tolerance analysis, and testing by the developer are required to validate their design.

14.2.3 Valve Misalignment

It is important that the solution developed ensures that valve misalignment does not exceed the requirements in the Section 9.1 of the *Open Rack V3 Blind Mate Quick Connector Specification*. Contributors to misalignment include, but are not limited to, gaps between subassemblies and components, skew between components due to gaps,

interface tolerances, manufacturing tolerances, component and sub assembly tolerances, deflections due to weight loading IT gear and support brackets, chassis friction, deflections from fluid and valve forces, and CTE (coefficient of thermal expansion) effects as the ambient and or fluid temperatures change. Appropriate structural analysis, tolerance analysis, and testing by the developer are required to validate their design.

14.3 Manifold Liquid Cooling Mounting Kit

The blind mate manifolds are mounted to the ORv3 frame using a Blind Mate Liquid Cooling Kit, which is included in the reference 3D CAD. Figure 14.3.1 highlights the included components:

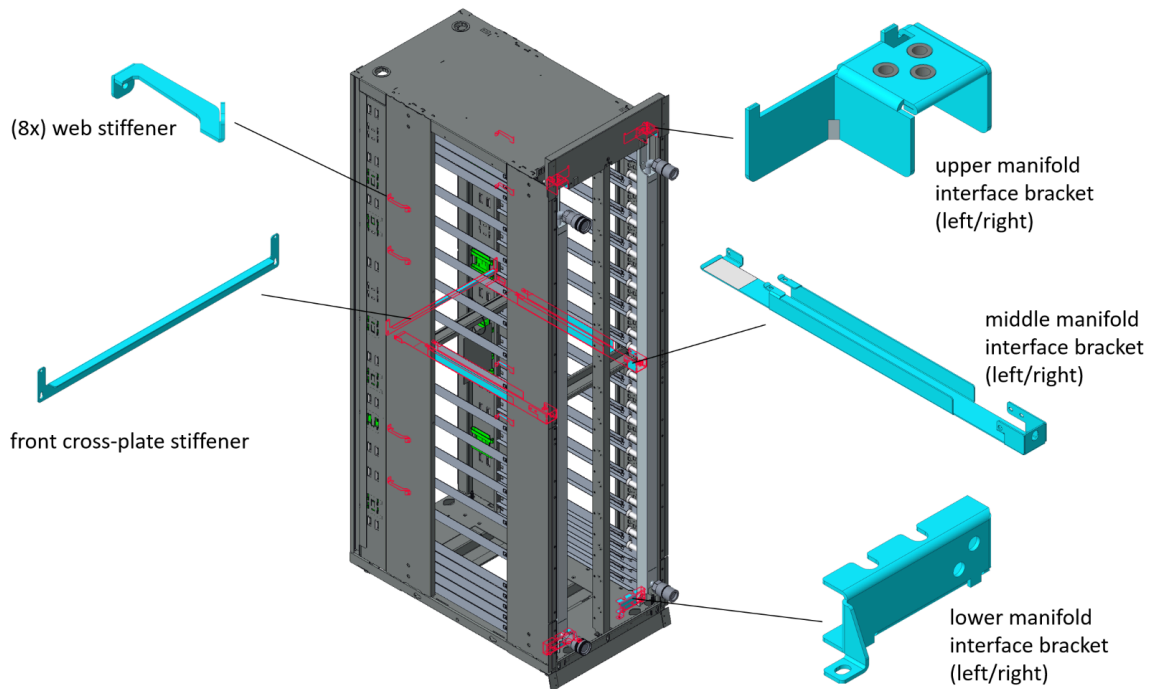


Fig. 14.3.1 Blind Mate Liquid Cooling Mounting Kit components

Description of liquid cooling kit components:

- **(8x) Web Stiffeners** – these mount into the front vertical members of the rack and provide additional stiffness to rack Datum A features which must counteract the forces from the blind mate IT gear injectors.
- **(1x) Cross-Plate Stiffener** – mounts to the front of the rack and provides additional frame stiffness to counteract the additional liquid cooling forces; the cross-plate is mounted in the same OU slot as the standard frame cross-braces.

- **Left/Right Lower Manifold Interface Brackets** – these serve as the lower mounting interface for the manifold Datum A surface (zone A1) and provide some additional structural support to help counteract the valve & fluid forces trying to push the manifold rearward,
- **Left/Right Middle Manifold interface Brackets** – these serve as the middle mounting interface for the manifold Datum A surface (zone A3) and provide additional structural support to help counteract the valve & fluid forces trying to push the manifold rearward; during installation into the rack, the bracket is registered off rack Datum A to minimize positional tolerance in the depth direction (Z-axis); the middle interface brackets are mounted to the standard ORv3 frame cross-braces (front & rear) and utilize the same OU slot within the rack; in the reference design, these braces are positioned in slot OU23, but could be +/- 1OU in either direction depending on end user needs. Other brace locations are possible; however, the developer will need to validate the design and will require design changes to some components, frame features, interfaces, and manifolds.
- **Left/right upper manifold interface brackets** – these serve as the upper mounting interface for the manifold Datum A surface (zone A2), provide some additional structural support to help counteract the valve & fluid forces trying to push the manifold rearward, and provide left/right constraint for the upper portion of the manifold via a guide pin in the bracket.
- Various fasteners to attach items listed above to the rack, see provided 3D CAD and 2D drawing for additional information.

14.4 Mechanical Attachment, Manifold to Rack

14.4.1 Upper interface fasteners

The manifold is attached to the upper rack interface bracket using (2x) M8 coarse thread-locking nuts installed from the rear of the rack. Recommended torque = 15 ± 2 N-m

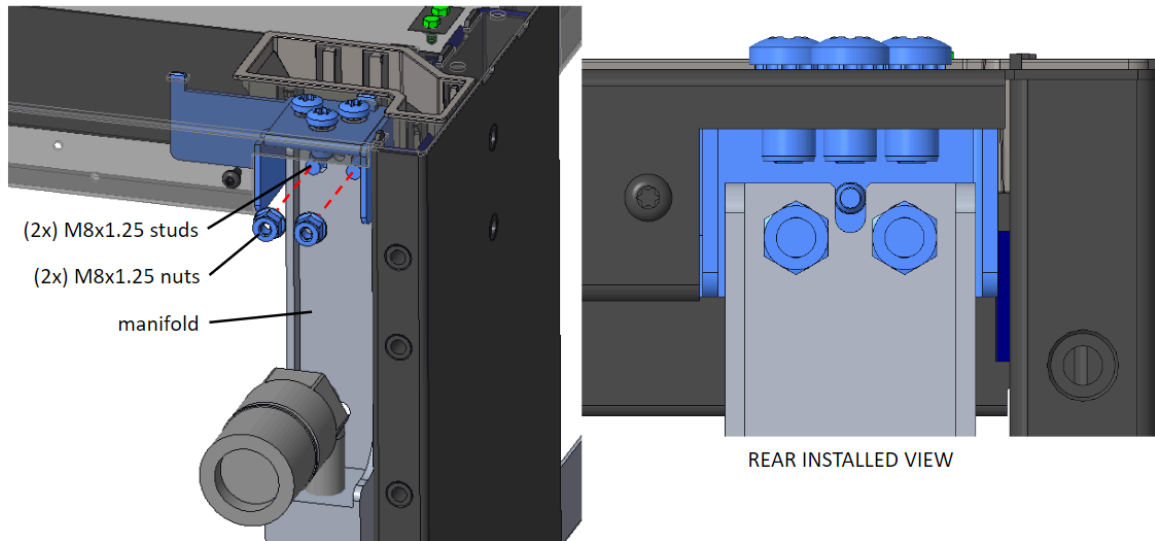


Fig. 14.4.1.1 - upper interface fasteners

14.4.2 Lower interface fasteners

The manifold is attached to the lower rack interface bracket using (2x) M8 coarse thread-locking nuts installed from the rear of the rack. In the reference design the M8 studs on the rack interface are mounted to a secondary stud plate, which can easily be replaced if damaged. Recommended torque = 15 ± 2 N-m

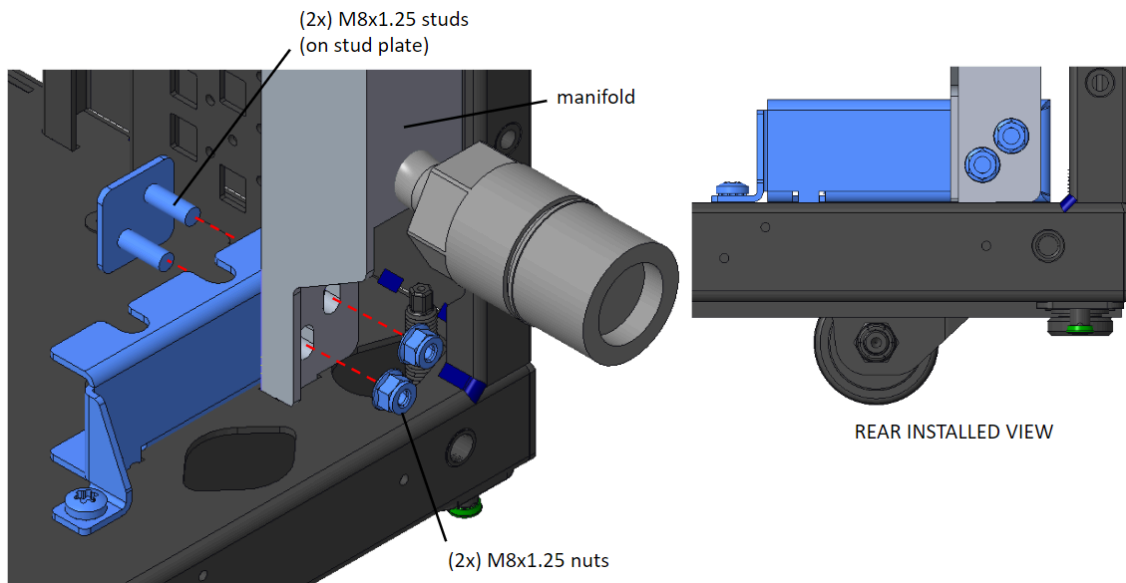


Fig. 14.4.2.1 - lower interface fasteners

14.4.3 Middle interface fasteners

The manifold is attached to the middle rack interface bracket using a single M12 coarse thread-locking nut installed from the front of the rack. Recommended torque = 30 ± 5 N-m

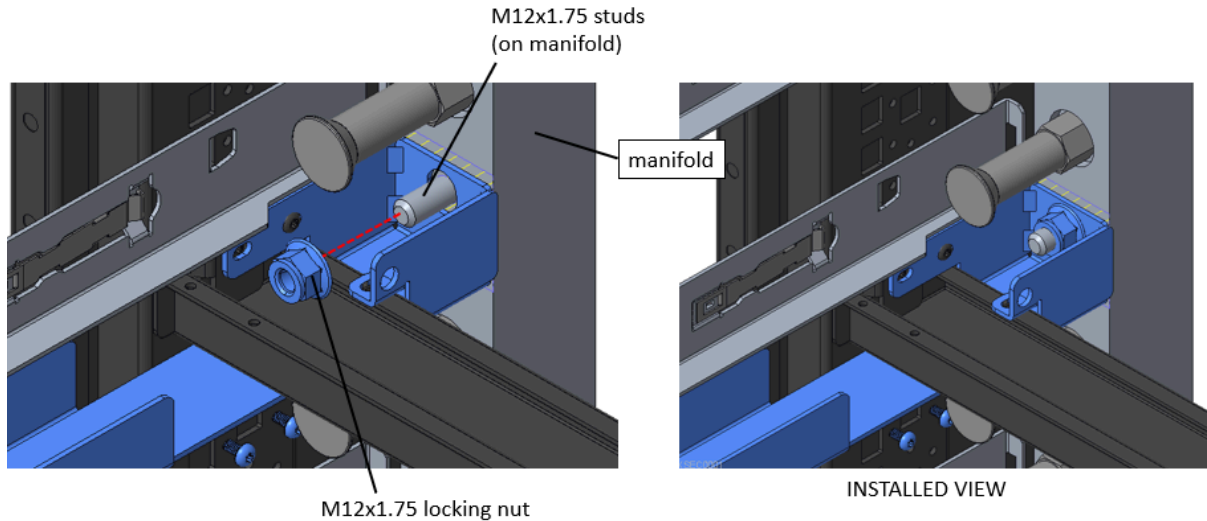


Fig. 14.4.3.1 - middle interface fasteners

14.5 IT Gear Design Limitations

There are a few restrictions in IT gear locations when using the blind mate manifolds.

These restrictions include:

- Cross-Brace location must be within OU22 - OU24 position
- Cross-Brace and Front Cross-plate Stiffener Bracket takes up 10U of space (both occupy the same OU slot position)

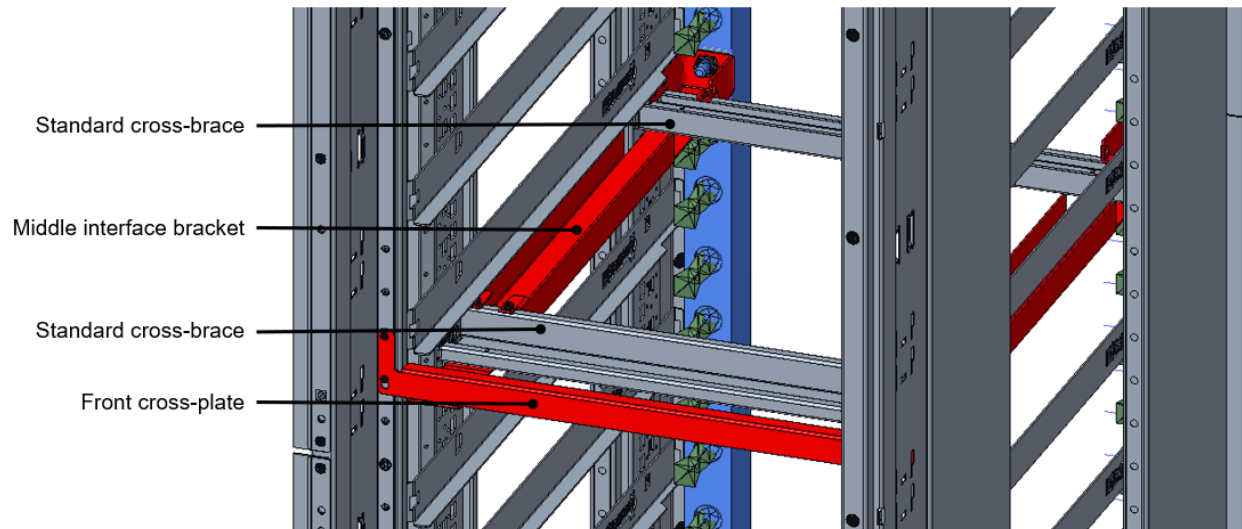


Fig. 14.5.1 - cross plate and middle interface brackets occupy same OU slot as standard cross-braces

- In the reference design there are no liquid cooling chassis connections (BMQC valves) in locations ou42 - ou44 due to manifold construction and mounting and power shelf power cable routing. If a developer wishes to add connections in these locations, the developer will need to consider impact to space for air purge and or fluid connections for supply/return.
- In the reference design there are no liquid cooling chassis connections (BMQC valves) in locations ou1 - ou4 due to manifold construction and mounting. If a developer wishes to add connections in these locations, the developer will need to consider the impact to leveling feet access and fluid connections for supply/return.
- OU1 slot position does not allow for full depth IT gear or IT gear brackets due to lower manifold interface brackets. See Figure below showing lower manifold brackets installed in the rack. The developer is free to determine how to use this space.

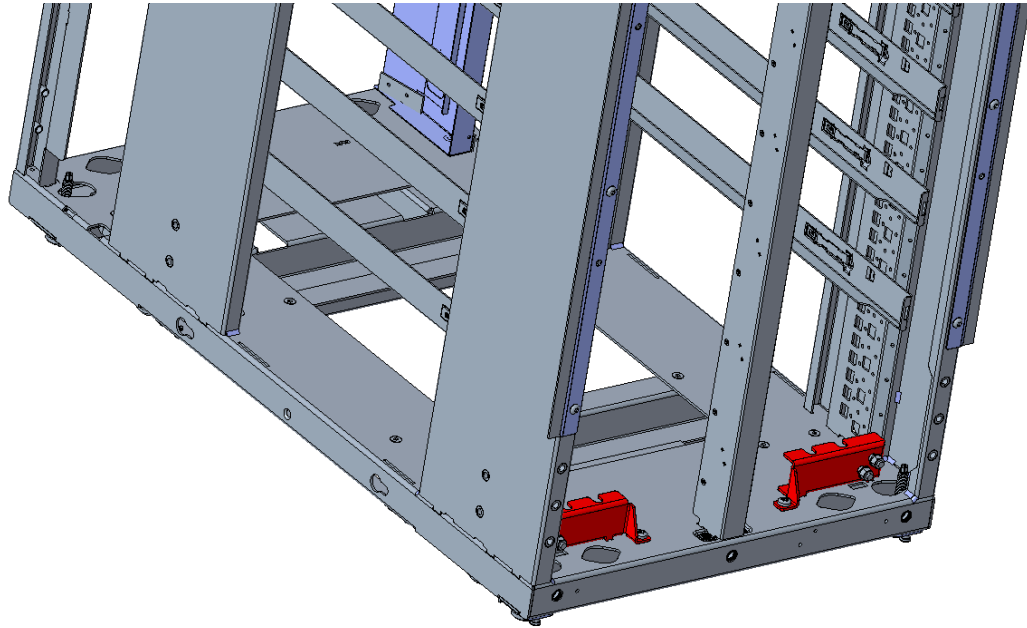


Figure 14.5.2 - Rear ISO View of the ORv3 Frame -
Lower Manifold Brackets Installed (in red)

14.6 Rack Cross Brace Location

The Open Rack V3 frame design supports a cross brace location between OU18 to OU27. However, when using the blind mate manifold the cross brace should be positioned within OU22-OU24 which is roughly in the center payload zone of the rack. Structural analysis of the reference design was completed with the cross-brace in position OU23. For other OU positions, additional FEA should be completed (both at the manifold and rack level) to ensure design goals for deflection & strength are achieved for the end application. The position of the liquid cooling mounting kit front cross plate and mid-interface brackets were designed to be used in the same position as the rack cross-brace.

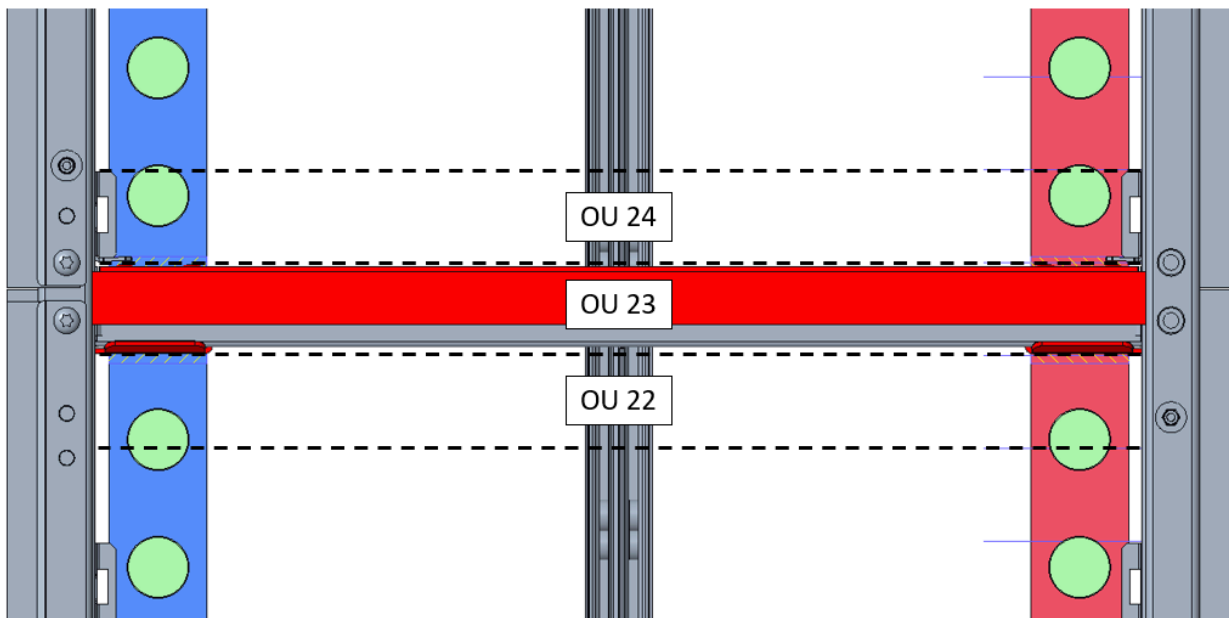


Fig. 14.6.1 - Cross-brace location options for liquid cooling configurations

14.7 Key Dimensions

A 3D model of the rack with IT Gear reference design, manifold reference design, and liquid cooling kit are provided with this document. A 2-D drawing of the key interface dimensions and tolerances is also provided.

14.8 Port locations

See the *Open Rack V3 Blind Mate Manifold Base Specification* for details on port locations and limitations.

14.9 Loading & Force considerations

Below, are some of the areas to consider with respect to the loads and forces in a blind mate rack solution:

- Forces from valves and fluid pressure
- Coefficient of thermal expansion affects – air or fluid temperatures influencing this.
- Weight of IT gear and components
- Total payload of the solution should not exceed ORv3 payload specifications.
- Transportation and handling

14.10 Allowable Frame Deflection

To meet the minimum design goals of the overall liquid cooling solution, which considers BMQC working range and backplane connector wipe, the rack, IT gear, and manifold have allowable budgets for deflections. The allowable rack frame deflections at the Datum A flange at any point shall be no more than 0.6mm (in the depth direction). The allowable deflection of the manifolds when mounted with the liquid cooling kit in ORv3 is defined in the manifold specification. The allowable deflection of the IT gear when mounted in the ORv3 is provided in the Open Rack V3 IT Gear Design Guide. The deflection maximums of the three components make up the total deflection budget for the liquid rack assembly with gear. These values along with the provided tolerances for each component are the variables to analyze and ensure the solution maintains the mating distances of the BMQC's within their working range. More details about this are discussed in Section 14.2.2.

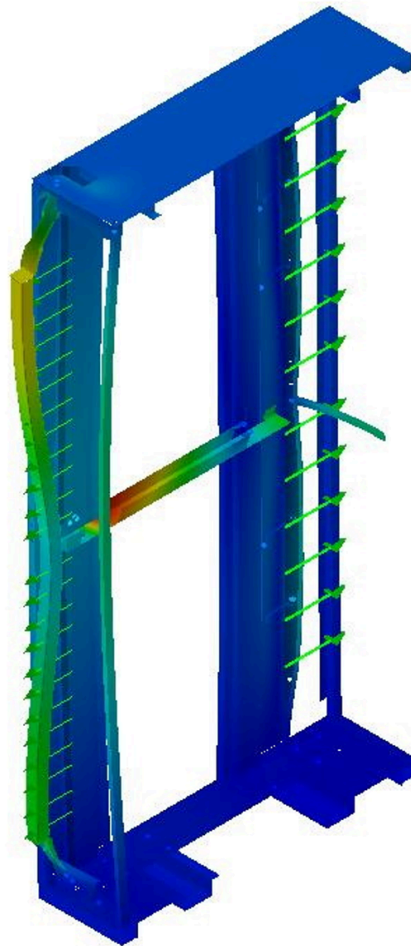


Fig. 14.10.1 - Example FEA of manifold and rack deflections (depth direction) resulting from hydraulic forces

Developers will also want to consider the effects of thermal expansion on manifolds & rack at elevated temperatures when mounted in the rack. To ensure the blind mate valves are within their allowable working range and misalignment, it is important that the developer consider the effects of manifold CTE, manifold deflection budget & tolerances, IT gear deflection budget & tolerances, rack deflection and tolerances, rack assembly tolerances, and external environmental inputs (shipping and handling, vibration, etc..). The tolerance requirements for the rack interfaces for the manifolds are defined in this specification. The blind mate valve specification provides the allowable misalignment and working range requirements.

14.11 Blind Mate Rack Environmental Requirements

- Operating Temperature¹: 10°C to 60°C* (Fluid Temperature 30C - 60C)
 - Operating Humidity¹: 85%max, 5°C dew point minimum
 - Non Operating Temperature¹: 5C to 65C
 - Non Operating Humidity¹: 5-95%
- ¹with Dowfrost LC25

14.12 Blind Mate Liquid Cooling Safety Requirements

14.12.1 General safety construction requirements

Any liquid cooling related components shall not contain any hazardous, flammable, or toxic substances classified by law or local regulation, and MSDS (Material Safety Data Sheet) must be provided. No liquid spillage is allowed during normal and abnormal operating conditions.

14.12.2 Plastic parts flammability

Minimum flammability requirements for manifold assembly components should be considered for each unique application but are not mandated in this specification.

Tubing can be made of minimum HB75 class material if the thickness of the material is < 3mm, or minimum HB40 class material if the thickness of the material is ≥ 3mm.

14.12.3 Hydrostatic pressure

The liquid cooled rack shall be designed to meet the following hydrostatic pressure and creep resistance tests.

Test items	Requirements	Reference	Criteria
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Hydrostatic pressure (Normal operation)	The following test shall be performed at normal operation room temperature. - the liquid cooled rack shall be designed to meet three times the maximum working pressure specified by the manufacturer. - ambient temperature: 45C - fluid temperature: 60C - fluid pressure: 3 x 50 psig = 150 psig	UL62368-1	During and after tests, there shall be no rupture, no leaks and no loosening of any connection or part.
Hydrostatic pressure (Abnormal operation)	The following test shall be performed at abnormal operating condition when single fault condition is applied to various liquid cooling components in the rack. - liquid cooled rack shall be designed to meet two times the maximum working pressure specified by the manufacturer at the maximum temperature measured during application of abnormal condition (such as covering the ventilations of the final end product- rack) - ambient temperature: 55C - fluid temperature: 60C - fluid pressure: 2 x 50 psig = 100 psig	UL62368-1	During and after tests, there shall be no rupture, no leaks and no loosening of any connection or part.

Creep resistance	Non-metallic liquid cooled components shall show no sign of deterioration such as cracking and embrittlement during/ after the following tests. - liquid cooled components or liquid cooled rack shall be conditioned for 14 days at a temperature of 87°C - Following the conditioning, the system shall comply with the above hydrostatic pressure tests	UL62368-1	Non- metallic parts shall show no sign of deterioration such as cracking and embrittlement
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14.12.4 Tensile requirement

Tubing and associated fittings that may be used in Manifold design shall meet the following tensile strength tests.

Test items	Requirements	Reference	Criteria
Tensile strength	The following tensile tests shall be performed on ten specimens of tubing material in accordance with ISO 527 standard. - A component level, five specimens of the various liquid cooled components shall be tested in the as received condition - An additional five specimens shall be tested after a conditioning test for 40 days at 38°C in a full draft air-circulating oven or in a water bath filled with the intended liquid and maintained at 38°C.	ISO 527 UL62368-1	The tensile strength after conditioning shall not be less than 60% of the tensile strength before the tests.

14.12.5 Mechanical strength

Test items	Requirements	Reference	Criteria
Force test	A force of $30\text{N} \pm 3\text{N}$ is applied to any accessible part for a duration of 5 sec. applied to any tubing or liquid cooled connections	UL62368-1	After tests, there shall be no rupture, no leaks and no loosening of any connection or part.

15. References

- [1] Open Rack V3 Base Specification
- [2] Open Rack V3 Blind Mate Quick Connector Specification (BMQC)
- [3] Open Rack V3 Blind Mate Manifold Base Specification
- [4] Open Rack V3 IT Gear Design Guide